

sliding window Protocol

AIM:

To write sliding window protocol using Python.

Create a sender program:

- (i) create a frame with fields [Frame no, Data]
- (ii) send the frames
- (iii) wait for acknowledgement.
- (iv) check Acknowledgement number.
- (v) set new frames.

Create a Receiver file:

- (i) Read a file called sender-Buffer
- (ii) Check the Frame no.
- (iii) If frame no accepted, write ACK no in receiver-Buffer.

Code:

sender.py

```
import time
import os
ws = int(input("window size"))
msg = input("enter message: ")
frames = [(i+1, ch) for i, ch in enumerate(msg)]
i = 0
while i < len(frames):
    window = frames[i:i+ws]
    with open("sender-Buffer:", "w") as f:
```

```

for no in window:
    f.write(f"{no}, {data}\n")
print("\n sending :", window)
time.sleep(2)
while not os.path.exists("Receiver-Buffer"):
    time.sleep(1)
with open("Receiver-Buffer") as f:
    ack, no, f.read().strip().split(',')
    no = int(no)
print("Received", ack, no)
if (ack == "ACK" and no == window[-1][0]):
    i += ws
elif ack == "NACK":
    i = no - 1
open("Receiver-Buffer", "w").close()
print("\n Transmission Completed")

```

receiver.py

```

import time, os
expected = 1
while True:
    if not os.path.exists("sender-Buffer"):
        time.sleep(1)
        continue
    with open("sender-Buffer") as f:
        frames = [(int(x.split(",")[0]).x.split(
            ",")[1].strip()) for x in f if x.strip()]
    if not frames:
        time.sleep(1)
        continue
    print("\n Received", frames)
    for no in frames:

```

if no! = expected :

with open ("Receiver Buffer ", "w") as f:
f.write (f" NACK. {expected} ")

break

expected += 1

else

with open ("Receiver Buffer ", "w") as f:
f.write (f"ACK, {frame [-1][0]}")

time.sleep()

Sender Terminal

window size : 3 Enter message: Angel

sending [(1, 'v'), (2, 'i'), (3, '3')]

Received : ACK 3

sending [(4, 'h'), (5, 'u')]

Received ACK 4

Transmission Completed.

Receiver Terminal

Received : [(1, 'v'), (2, 'i'), (3, '3')]

Received : [(4, 'h'), (5, 'u')]

Result

Thus the code is implemented for sliding window.

Ek